

Dynamic Topology Optimization of Mega-Constellation Satellite Networks Based on Morphological Computing

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Abstract—The low Earth orbit (LEO) satellite constellation plays a very important role in the 6G Internet of Things (IoT). But the links of LEO mega-constellations satellite networks (MCSN) are interrupted frequently and the topology is highly dynamic, resulting to poor service quality for IoT. This paper proposes a morphological computing dynamic topology optimization (MCDTO) framework to address this issue. By constructing a spatiotemporal unit decomposition (STUD) model, the continuous time-varying network is decoupled into discrete satellite clusters. A dynamic satellite clustering boundary optimization algorithm is proposed and reduced the topology management complexity from $O(N^2)$ to $O(KN)$. Experiments show that the MCDTO method can improve the stability of critical links to 0.6639, and increase the standard deviation of cluster distribution from 2.1 to 13.2. The MCDTO method brings computational efficiency for dynamic routing and robust management of MCSN, significantly increases resource concentration and service resilience.

Keywords—*morphological computing, mega-constellation satellite network, spatiotemporal unit decomposition, topology optimization.*

I. INTRODUCTION

Driven by needs of the satellite Internet and the space integrated network, the MCSN, such as Starlink, are developing rapidly [1]. This type of network has the core characteristics of wide coverage, topology variability, and service differentiation[2], but its complex spatiotemporal dynamic characteristics pose a serious challenge to traditional network optimization methods: on the one hand, the frequent on-off of inter-satellite-links caused by high-speed satellite motion renders traditional static topology models ineffective[3]. On the other hand, the distributed resource management and global quality of service (QoS) guarantee of large-scale constellations are difficult to balance through local optimization strategies [4].

The current research on satellite network topology optimization mainly focuses on three directions: dynamic routing protocol design, virtual node modeling, and snapshot sequence optimization. For example, the topology control

based routing algorithm [5], reduces the frequency of topology updates through a virtual node strategy [6], while the Zhangs' team improved the applicability of Dijkstra's algorithm for time-varying topology [7]. However, existing methods still have obvious limitations: (1) Traditional time slice division methods can not effectively capture the spatio-temporal coupling laws of satellite network morphological evolution [8]. (2) The quantification of link stability is insufficient, and existing research mostly evaluates based on geometric distance or available link duration, without integrating multi-dimensional state parameters such as network load and business priority [9].

Morphological computing provides a new way to solving the above problems by extracting the morphological characteristics of the system such as spatial distribution patterns and dynamic evolution laws to build a computable models. Its core advantages lie in that it can achieve dimensionality reduction representation of high-dimensional dynamic systems [10]. This paper firstly introduces the theory of morphological computing into the field of topology optimization of MCSN, the contributions of this paper are summarized as: (1) **Combine morphological computing and MCSN to propose a MCDTO method.** A UCTD model has been proposed that can effectively describe the periodic and hierarchical distribution of satellite network coverage. (2) **Quantitative mechanism for link stability of inter-satellite-link in morphology perception.** We innovatively model the topology of satellite networks as a morphology, which enables the use of embodied intelligence theory to optimize the network topology. The method breaks through the traditional satellite topology evaluation bottleneck based on available link duration, and achieves an end-to-end delay reduction.

II. SYSTEM MODEL

This section is based on the theory of morphological computing to construct a dynamic topology optimization model for MCSN, mainly including STUD and link stability metric (LSM). The model architecture is shown in Fig.1.

In the scenario shown in Fig. 1, two parts of satellites in the mega-constellation are constructed into two satellite clusters, providing services to users in two regions. From the perspective of the entire satellite network, there is a significant difference in services between the two clusters. From the perspective of a single cluster, the users in this area are relatively concentrated, which is conducive to satellite clusters, mutual cooperation, sharing of satellite resources, and providing services for users.

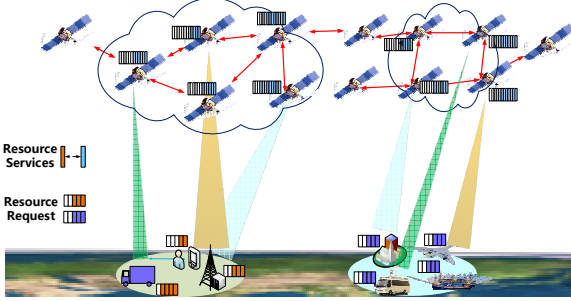


Fig. 1 Topology optimization scenario of satellite network based on STUCs

A. Spatiotemporal Unit Decomposition Model

Due to the large number of satellites in the LEO constellation and their high-speed movement, its network topology is very complex. This large-scale spatiotemporal variability leads to the failure of static network management methods. To address this challenge, the paper adopts the idea of STUC, and decomposes continuous time-varying topology into discrete STUC. It realizes more refined and flexible topology management. STUC decomposes the continuous time-varying process of satellite network topology into several STUC with stable morphological characteristics. In terms of time dimension, the relative position change of satellite nodes within the unit cluster does not exceed the threshold $\Delta d_{\max}$, which is determined by orbital periodicity and coverage overlap. In terms of spatial dimension, the inter-satellite-link connectivity within the unit cluster remains stable and less than number of link disconnections λ_{thres} .

The time-varying characteristics of satellite network topology are derived from the laws of orbital mechanics, and its periodicity can be accurately modeled through orbital parameters. Therefore, in order to decompose the satellite network and construct a spatiotemporal unit decomposition model, it is necessary to first extract features based on satellite orbital parameters: semi-major axis a , inclination i , ascending node right ascension Ω and coverage periodicity, and use morphological clustering algorithms to generate an initial set of STUCs $C = \{c_1, c_2, \dots, c_k\}$. Then, based on task requirements and dynamic topology changes, dynamic adjustments can be made. During the adjustment, the shape

similarity between adjacent unit clusters $S(c_i, c_j) = \frac{\|f_i - f_j\|}{\Delta t}$ (f as shape feature vectors) can be calculated in real time according to the satellite motion equation $\frac{dr}{dt} = \sqrt{\frac{\mu}{a^3}} v$ (μ as

the Earth's gravitational constant and r position vector)[11], and unit clusters can be merged or split to maintain stability.

B. Quantification of Link Stability for Shape Perception

In order to realize the fine routing decisions, it is necessary to establish a link stability evaluation system which is strongly coupled with network configuration. Traditional link quality indicators based on distance or delay are difficult to adapt to the spatiotemporal correlation characteristics of mega-constellations. The LSM function combines the track dynamics characteristics with network state parameters through multi-dimensional sensing mechanism to form a link evaluation standard with temporal and spatial adaptability. For inter-satellite-links l_{ij} , the stability weight ω_{ij} is jointly determined by geometric parameters and network state:

$$\omega_{ij} = \alpha \cdot \frac{\theta_{ij}(t)}{\theta_{\max}} + \beta \cdot \frac{\dot{d}_{ij}(t)}{\dot{d}_{\max}} + \gamma \cdot \frac{L_{ij}(t)}{L_{\max}} + \delta \cdot \frac{\Delta \tau_{ij}(t)}{\Delta \tau_{\max}} \quad (1)$$

Among them, $\theta_{ij}(t)$ is the elevation angle of the link, reflecting the visibility between satellites. And $\dot{d}_{ij}(t)$ is the rate of change in distance, which characterizes the lifespan of the link. $L_{ij}(t)$ is the link load, dynamically adjusted according to the queue length. For latency jitter $\Delta \tau_{ij}$, it is calculated from historical transmission packets. For the weight coefficient $\alpha, \beta, \gamma, \delta$, satisfy $\alpha + \beta + \gamma + \delta = 1$.

Due to the dynamic time-varying nature of MCSN, a link weight dynamic update mechanism is also been proposed to update the characteristic changes of MCSN in network topology. The link state parameters within the STUC will be collected periodically. By adjusting the weight coefficients through a fuzzy logic controller, priority is given to ensuring high priority services, such as emergency telemetry data. The weight coefficients $\alpha, \beta, \gamma, \delta$ is adjusted within each STUC. The input variables are business priority P and network congestion index C . By using STUC, the complexity of the original dynamic optimization problem is reduced from $O(N^3T)$ to $O(K \cdot (N/K)^3)$, where $K = \lceil \sqrt{N} \rceil$ is the number of unit clusters.

III. EXPERIMENT

The experiment is based on the joint simulation platform of MATLAB and STK, and generates dynamic topology data of LEO satellite constellations through STK. The experiments cover satellite motion trajectories with orbit heights of 550-1200km and inclinations of 53°-98°. And the experiments set different satellite scales and introduced Poisson distributed burst traffic and 5% -15% random node failure attacks to test the adaptability of the topology optimization framework.

The inter-satellite-links are the key factor affecting satellite clusters and satellite services. We analyzed the link stability before and after the establishment of the satellite cluster. We use color depth and numerical labeling to show the link stability between different satellite clusters. The colors closer to yellow indicating more stable links. As can be

seen from the Fig. 3, most areas are displayed in dark blue, indicating low link stability, while specific cells display higher stability values, such as 0.6633 and 0.6504. These numerical values show that the optimized network has high stability on specific links, which reflects the effectiveness of morphological calculation methods in improving the topological stability of satellite network.

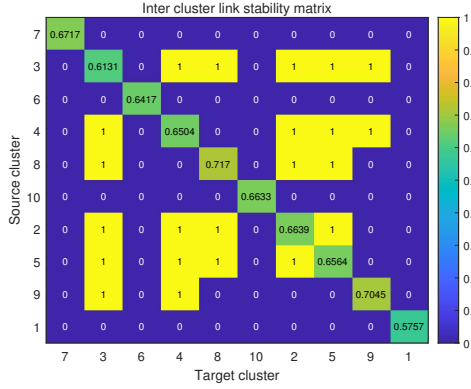


Fig. 2 Stability matrix of inter-satellite-links

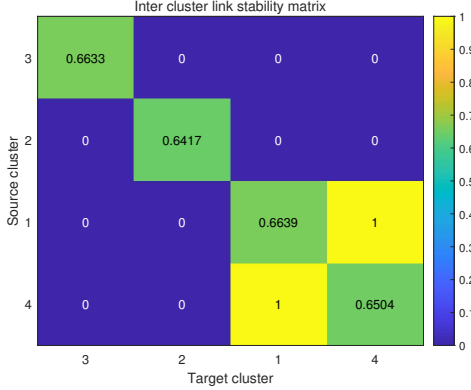


Fig. 3 Stability matrix of inter satellite cluster links

The Fig. 4 shows the results of optimizing the size distribution of satellite clusters. The blue bars represents before optimization and orange bars represents after optimization. The cluster size distribution before optimization is relatively uniform, and the standard deviation of $\sigma=2.1$, indicating that the number of satellites is evenly distributed among the STUC. After optimization, the cluster size distribution becomes uneven, and the standard deviation $\sigma=13.2$, with a difference of nearly 6 times. This change may reflect the priority of key clusters in the optimization process, so as to improve network connectivity and increase coverage by clustering. Optimizing the uneven distribution of cluster size shows that the morphological calculation methods is effective in adjusting the topology structure of satellite network, ensuring resource concentration and performance.

IV. CONCLUSION

By constructing STUD model and LSM, the problem of strong topological dynamics of MCSN and the failure of traditional static methods are effectively solved. STUD discretizes continuous time-varying topology into a morphologically stable STUC. Through the mechanism of orbit parameter clustering and dynamic adjustment, the complexity of topology optimization is reduced and the

calculation efficiency is significantly improved. LSM integrates track dynamics and network state, and combines with a fuzzy logic controller to dynamically adjust weight coefficients, optimizing link stability, and ensure high-priority services. The experimental results show that the stability of critical links has been improved. And the coverage and flexibility are enhanced, which verifies the effectiveness of morphological computing methods in dynamic topology optimization. This paper provides theoretical support for robust management and dynamic routing of MCSN, and engineering practicality and academic innovation.

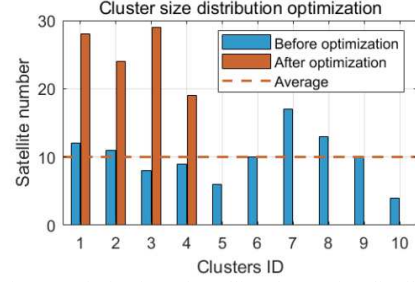


Fig. 4 Optimization of satellite cluster size distribution

ACKNOWLEDGMENT

This work is supported by Research Program of National University of Defense Technology under Grant No. ZK24-59.

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